LINUX PRIVILEGE ESCALATION AUTOMATION TOOLKIT

Project Documentation

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1. INTRODUCTION

This project focuses on developing an automated toolkit that scans Linux systems to detect privilege escalation vulnerabilities. Privilege escalation allows a normal user to gain root/admin access due to system misconfigurations. The toolkit automates scanning tasks and provides structured output for analysis.

2. OBJECTIVES

• Automate privilege escalation detection

• Identify system misconfigurations

• Generate security reports

• Provide practical cybersecurity experience

3. METHODOLOGY

The toolkit follows these steps:
1. System Information Collection
2. SUID File Scanning
3. Permission Analysis
4. Cron Job Detection
5. Output Analysis

4. IMPLEMENTATION WITH SCREENSHOTS

STEP 1: Environment Setup

System updated, Python installed, project folder and files created.

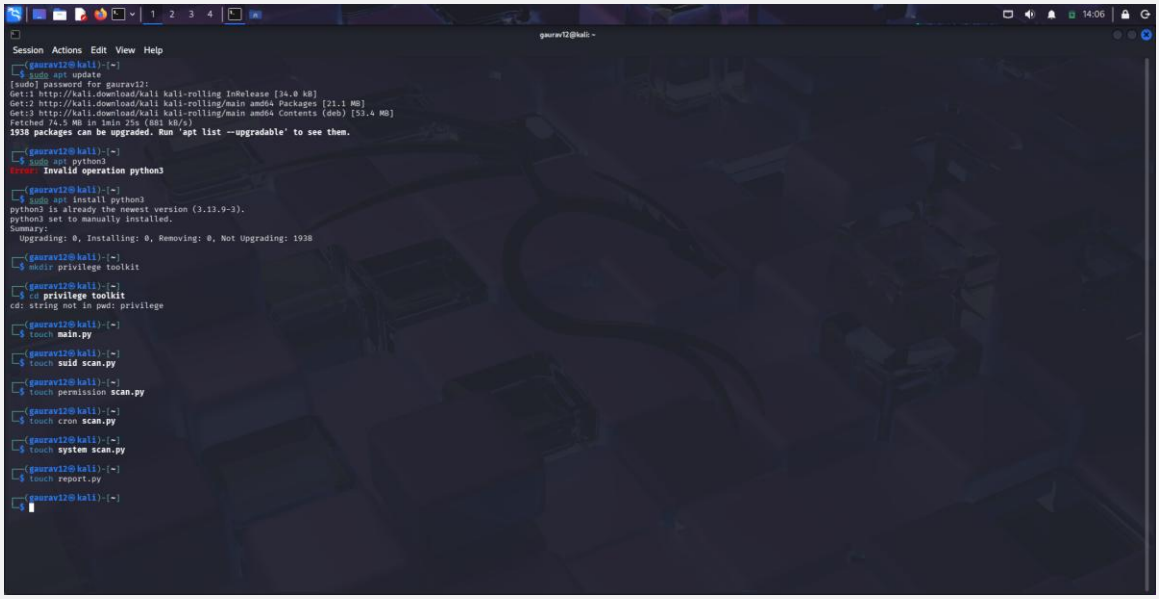

Figure 1: Environment setup and file creation

STEP 2: System Scan Code

Code for collecting system information (whoami, id, uname).

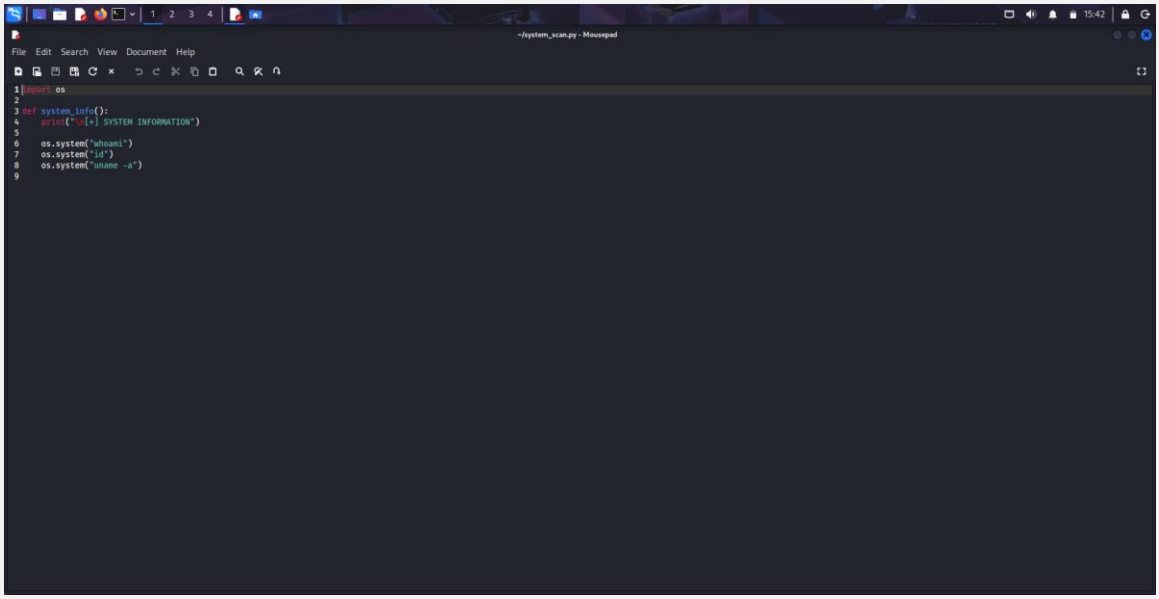

Figure 2: system_scan.py code

STEP 3: SUID Scan Code

Code for scanning SUID binaries using find command.

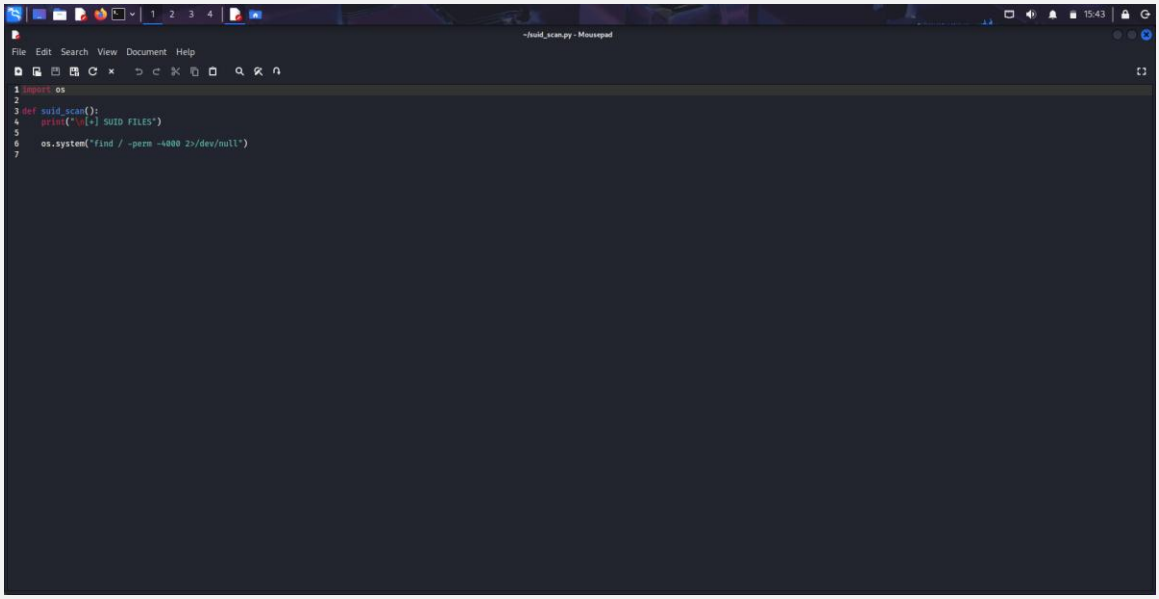

Figure 3: suid_scan.py code

STEP 4: Cron Job Scan Code

Code for listing cron jobs and scheduled tasks.

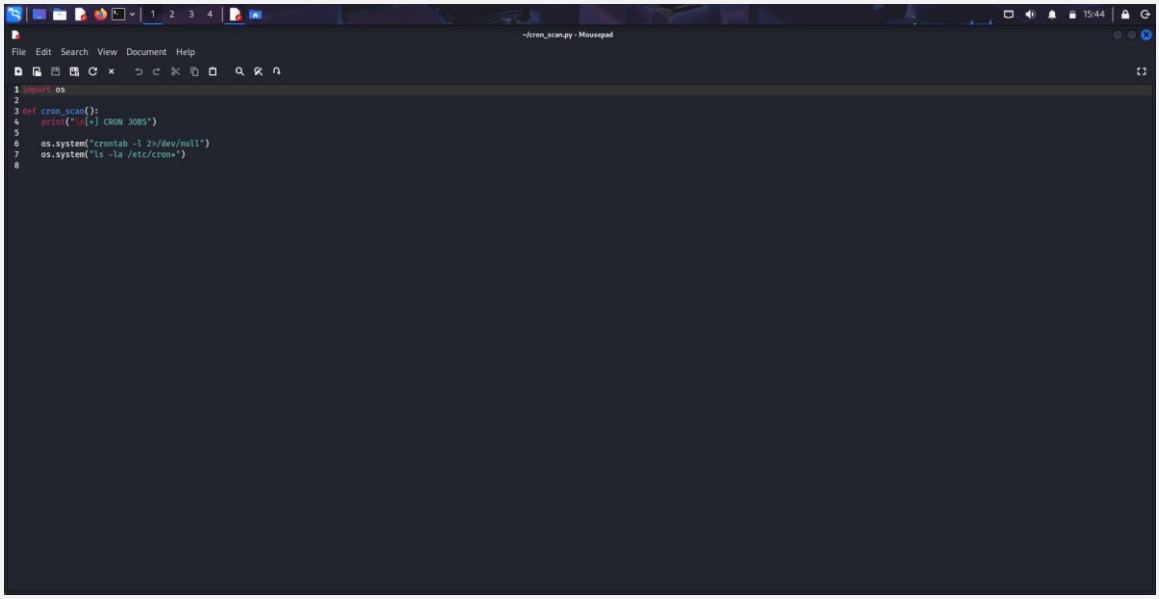

Figure 4: cron_scan.py code

STEP 5: Main Program Code

Main file combining all modules and running scans.

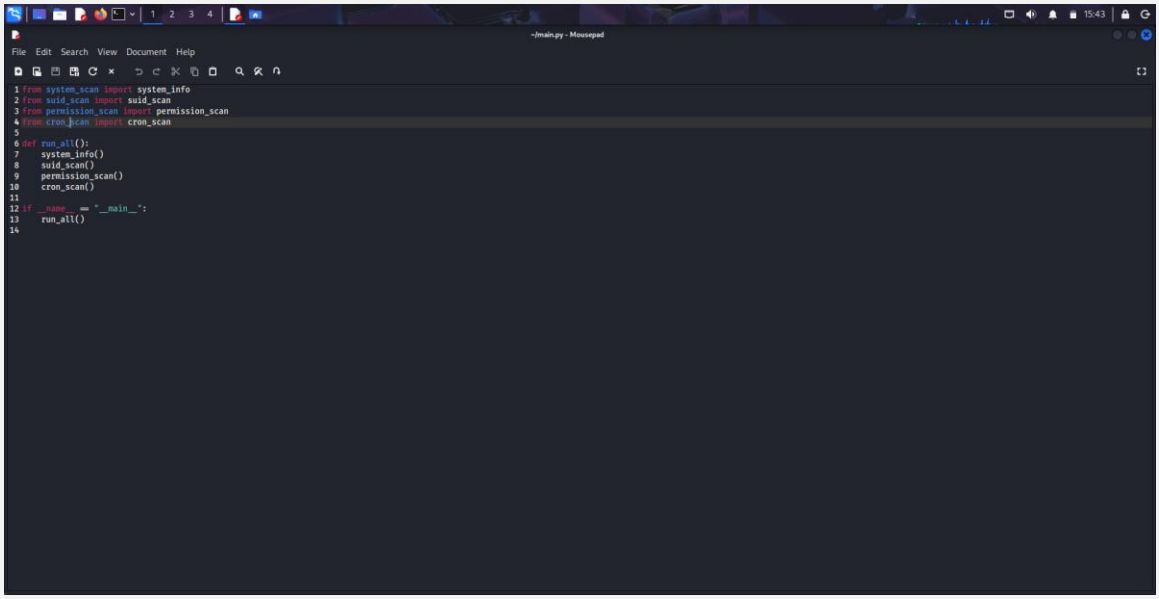

Figure 5: main.py code

STEP 6: Execution Output

System Information Output:

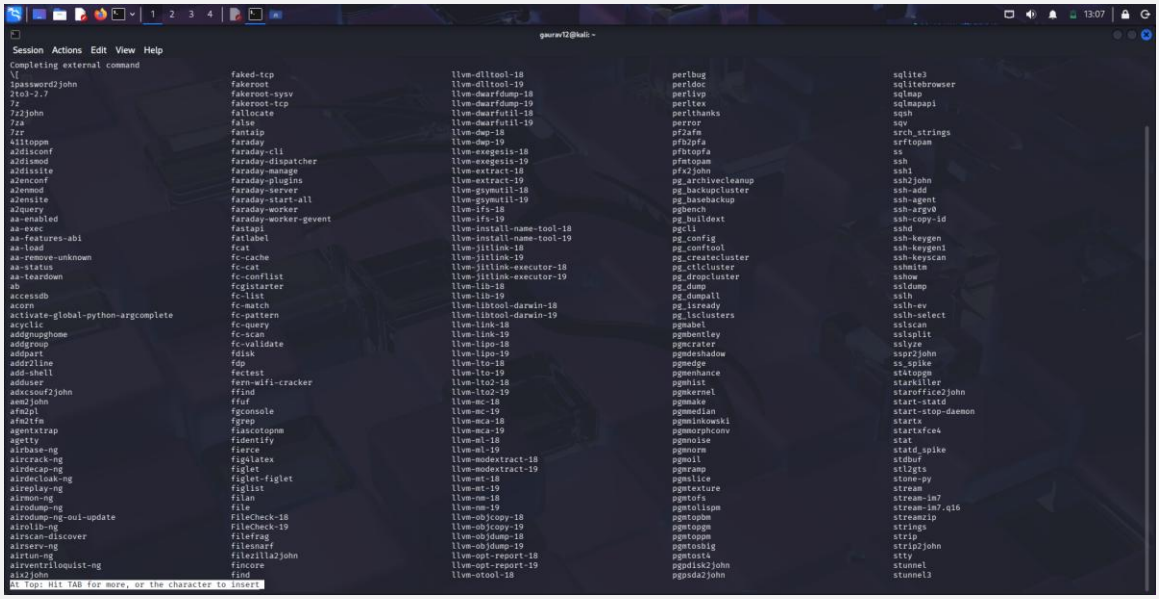

Figure 6: System info output

Toolkit Execution Output:

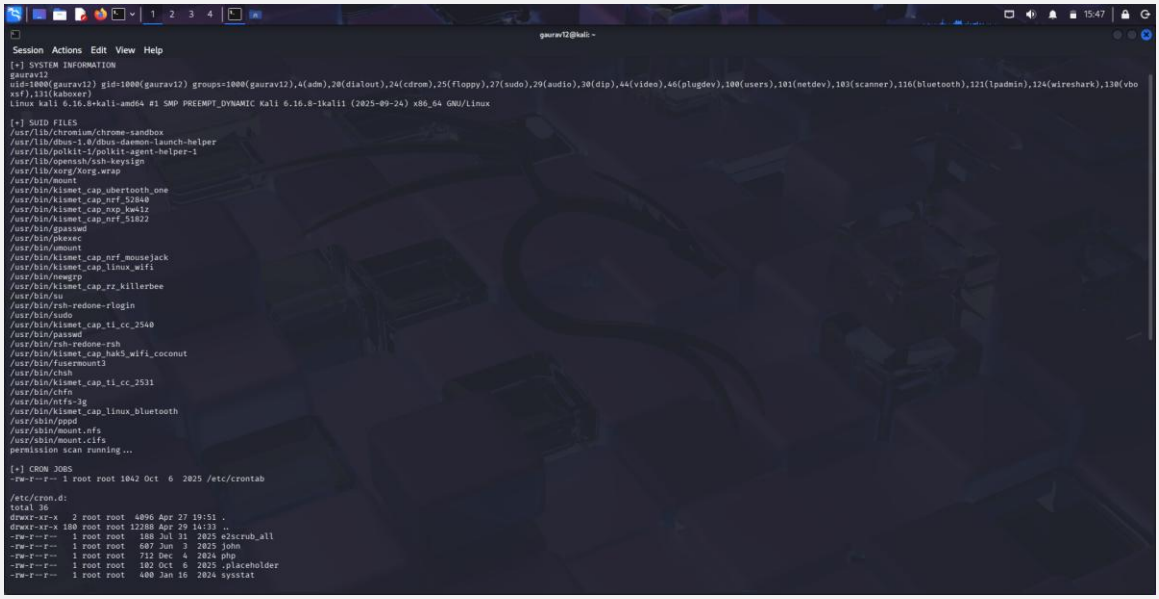

Figure 7: Toolkit run output

5. RESULTS

The toolkit successfully identified SUID binaries, detected cron jobs, displayed system information, and automated scanning tasks.

6. CONCLUSION

This project demonstrates automation in cybersecurity scanning. It provides practical exposure to privilege escalation detection and system auditing.